Sahaj Bhadja

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Summary

Embedded-leaning software engineer with 2+ years building **C++/Python systems**, **Linux-based automation tooling**, and **hardware-adjacent integration workflows** with strong debugging, reliability, and cross-functional collaboration in fast iterations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Web Technologies, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **fault-tolerant FastAPI services** and asynchronous job orchestration for 5K+ monthly workflows, adding validation, retry, and state-tracking logic that improved runtime robustness and failure recovery behavior in production.
- Diagnosed and resolved system-level performance bottlenecks across data and service boundaries, reducing latency from **1.2s to 350ms** for 500+ users through targeted query and interface optimization.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **Python/Java microservice workflows** with AWS Glue and S3, improving response times by **30%** while strengthening reliability of cross-system communication and backend processing under distributed deployment conditions.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, improving CI efficiency by **68%** and reducing technical debt by **20%** through continuous defect prevention practices.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed high-throughput **Python and TypeScript-backed modules** handling 10K+ daily requests, building robust API interfaces and structured processing logic while maintaining system correctness under fast-changing product requirements.
- Built operational tooling and CI/CD checks to speed delivery, reducing turnaround time by **60%** while improving release confidence and issue traceability across production and support workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented REST/WebSocket components supporting eight workflows and 2M+ synchronized records, improving real-time state consistency and reducing system-level communication issues across integrated application services.
- Automated Docker/Jenkins build-test-release loops, reducing deployment time by **50%** while improving reproducibility and enabling faster debug-fix cycles for cross-functional engineering and product teams.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Designed Python-based evaluation pipelines over 100K+ trajectories to benchmark behavior and isolate failure modes, improving reproducibility and strengthening engineering confidence during iterative development of complex systems.
- Partnered with multidisciplinary engineers to investigate root causes across algorithm and implementation layers, translating diagnostics into actionable fixes and clearer technical documentation for downstream development work.

Skills

Languages: **C++**, **Python**, **C** exposure, Java, Bash scripting, SQL
Embedded/Systems: Linux development, interface-level debugging, automation tooling, performance tuning, OOP design
Engineering: FastAPI, Flask, REST APIs, microservices, unit/integration testing, CI/CD, SCM workflows
Tooling: Docker, Jenkins, Git, AWS (S3/Glue), observability-driven troubleshooting

Projects

PiXMod | *Python, Next.js, MongoDB, Transformers*

- Built an AI-powered image processing platform with segmentation and retrieval modules, processing 1K+ assets while improving algorithm performance, pipeline reliability, and maintainability of production-serving service components.

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with interactive Gantt visualization, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ simulations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed a schema-aware SQL workspace for 10+ databases with structured validation and execution controls, improving correctness and reducing troubleshooting effort for complex data operations in engineering workflows.